SVKM's NMIMSSVKM's NMIMS
School of Technology Management & Engineering, Chandigarh
A.Y. 2026 - 27
Course: Database Management Systems

Project Report

Program	Masters in computer applications	
Semester	I	
Name of the Project:	Library Management System	
Details of Project Members		
Batch	Roll No.	Name
2	A053	Afifa Shaikh
1	A023	Umam Hashmi
Date of Submission:		

Contribution of each project Members:

Roll No.	Name:	Contribution
A053	Afifa Shaikh	
A023	Umam Hashmi	

Github link of your project:

Note:

1. Create a readme file if you have multiple files
2. All files must be properly named (Example:R004_DBMSProject)
3. Submit all relevant files of your work (Report, all SQL files, Any other files)
4. **Plagiarism is highly discouraged (Your report will be checked for plagiarism)**

Rubrics for the Project evaluation:

First phase of evaluation: Innovative Ideas (5 Marks) Design and Partial implementation (5 Marks)	10 marks
Final phase of evaluation Implementation, presentation and viva, Self-Learning and Learning Beyond classroom	10 marks

Project Report

Library Management System

By-

Afifa Shaikh, Roll number: A053

Umam Hashmi, Roll number: A023

Course: DBMS

AY: 2026-27

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I. Storyline

A small college library lends books to students. At the moment all records are kept on paper, so it is hard to answer simple questions: Who has this book? When is it due? How many copies are left? Which books are overdue?

To fix this, we design a small database with three kinds of data:

- **Members**: students registered in the library.
- **Books**: the books the library owns, with the number of copies available.
- **Issues**: a record created every time a member borrows a book (issue date, due date, return date).

Requirements:

1. Store member details (name, email, phone, join date).
2. Store book details (title, author, category, copies available).
3. A member can borrow many books, and a book can be borrowed by many members over time.
4. Every borrowing must record the issue date, due date and return date (empty until returned).
5. The librarian should be able to find unreturned books, overdue books, popular books, and fines.

II. Components of Database Design

Entities, attributes and primary keys

Entity	Attributes	Primary Key
Member	member_id, name, email, phone, join_date	member_id
Book	book_id, title, author, category, copies_available	book_id
Issue	issue_id, member_id, book_id, issue_date, due_date, return_date	issue_id

Foreign keys: Issue.member_id references Member.member_id, and Issue.book_id references Book.book_id.

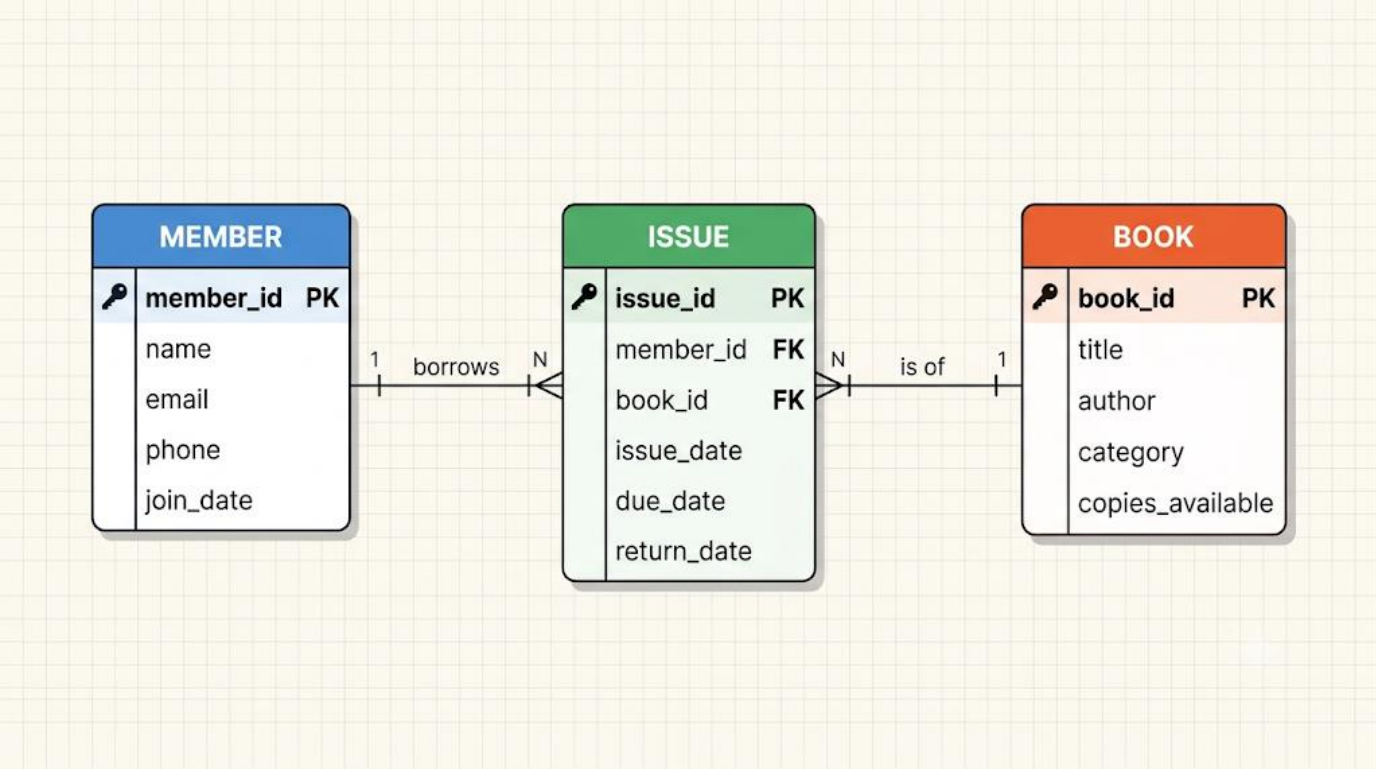
Relationships, cardinality and participation

Relationship	Cardinality	Participation
Member borrows Book (through Issue)	Many-to-Many (M:N)	Member: partial, Book: partial
Member has Issue	One-to-Many (1:N)	Member: partial (a member may never borrow), Issue: total (every issue belongs to a member)

Book appears in Issue	One-to-Many (1:N)	Book: partial (a book may never be issued), Issue: total (every issue is for a book)

In simple words: one member can have many issue records, one book can have many issue records, and every issue record must have exactly one member and one book. The Issue table turns the many-to-many relationship into two one-to-many relationships.

III. Entity Relationship Diagram



IV. Relational Model

The ER diagram converts into **3 tables**. Primary keys are marked PK and foreign keys FK:

- **Member**(**member_id** PK, name, email, phone, join_date)
- **Book**(**book_id** PK, title, author, category, copies_available)
- **Issue**(**issue_id** PK, member_id FK, book_id FK, issue_date, due_date, return_date)

Conversion rules used: each strong entity becomes a table; the one-to-many relationships are stored by placing the "one" side's primary key as a foreign key in the "many" side (Issue).

V. Normalization

Starting (unnormalized) table: one row per member, with several books in one cell.

member_id	member_name	phone	books_borrowed
1	Aarav Sharma	9876543210	Database Systems, Wings of Fire

1NF (First Normal Form): every cell must hold a single value. We give each borrowed book its own row.

Loan(member_id, member_name, phone, book_id, title, author, issue_date, due_date, return_date) with key (member_id, book_id, issue_date).

2NF (Second Normal Form): no partial dependency on part of a composite key.

- member_id → member_name, phone (depends only on member_id)
- book_id → title, author, category (depends only on book_id)

So we split into **Member**, **Book** and **Issue**.

3NF (Third Normal Form): no non-key column depends on another non-key column. In Member, Book and Issue every non-key column depends only on its table's key. (due_date is chosen by the librarian for each issue and can be extended, so it is not a fixed function of issue_date.)

BCNF (Boyce-Codd Normal Form): for every functional dependency X → Y, X must be a candidate key.

- Member: member_id → all columns (also email → all columns, since email is UNIQUE and is a candidate key)
- Book: book_id → all columns
- Issue: issue_id → all columns

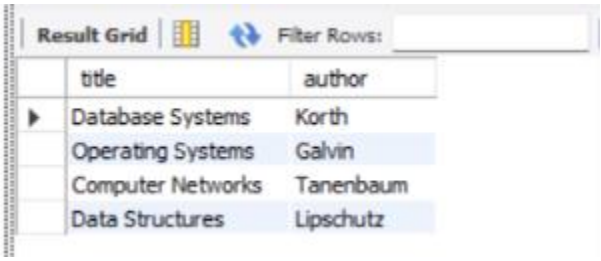
All determinants are candidate keys, so the database is in **BCNF**. There is no repeated data and no insert, update or delete anomaly.

VI. SQL Queries

Q1. (WHERE) List all Computer Science books.

SELECT title, author FROM Book WHERE category = 'Computer Science';

SCREENSHOT-

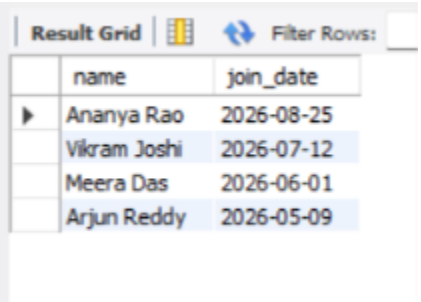


title	author
Database Systems	Korth
Operating Systems	Galvin
Computer Networks	Tanenbaum
Data Structures	Lipschutz

Q2. (ORDER BY) Members who joined after 1 May 2026, newest first.

SELECT name, join_date FROM Member
WHERE join_date > '2026-05-01' ORDER BY join_date DESC;

SCREENSHOT-



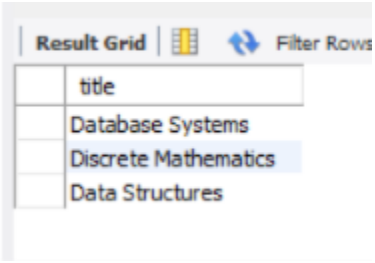
Result Grid | Filter Rows:

name	join_date
Ananya Rao	2026-08-25
Vikram Joshi	2026-07-12
Meera Das	2026-06-01
Arjun Reddy	2026-05-09

Q3. (LIKE) Books whose title starts with D.

SELECT title FROM Book WHERE title LIKE 'D%';

SCREENSHOT-



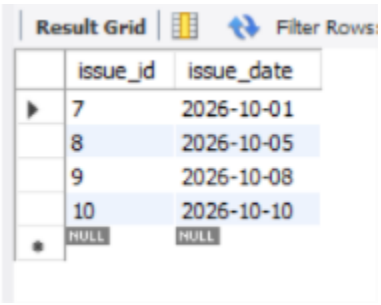
Result Grid | Filter Rows

title
Database Systems
Discrete Mathematics
Data Structures

Q4. (BETWEEN) Books issued between 1 Oct and 10 Oct 2026.

```
SELECT issue_id, issue_date FROM Issue
WHERE issue_date BETWEEN '2026-10-01' AND '2026-10-10';
```

SCREENSHOT-



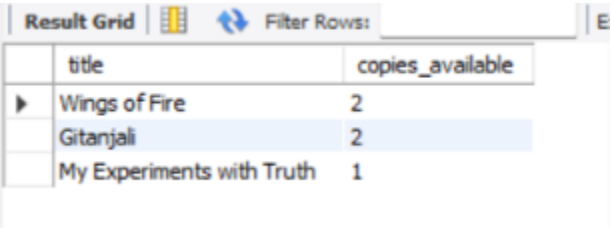
Result Grid | Filter Rows:

issue_id	issue_date
7	2026-10-01
8	2026-10-05
9	2026-10-08
10	2026-10-10
NULL	NULL

Q5. (IN) Biography and Poetry books with their copies.

```
SELECT title, copies_available FROM Book
WHERE category IN ('Biography','Poetry');
```

SCREENSHOT-



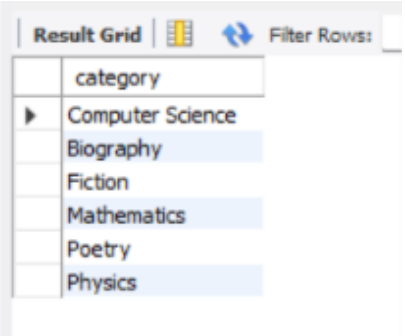
Result Grid | Filter Rows:

title	copies_available
Wings of Fire	2
Gitanjali	2
My Experiments with Truth	1

Q6. (DISTINCT) List the different book categories.

SELECT DISTINCT category FROM Book;

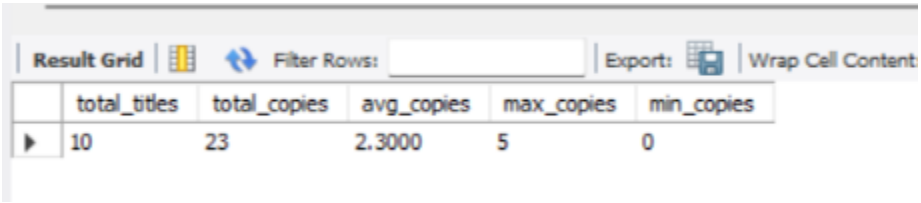
SCREENSHOT-



Q7. (Aggregate functions) Total books, total copies, average, maximum and minimum copies.

```
SELECT COUNT(*) AS total_titles, SUM(copies_available) AS total_copies,
      AVG(copies_available) AS avg_copies,
      MAX(copies_available) AS max_copies, MIN(copies_available) AS
min_copies
FROM Book;
```

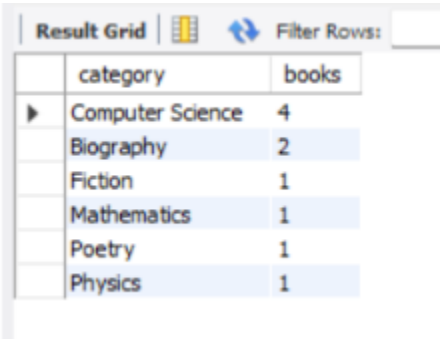
SCREENSHOT-



Q8. (GROUP BY) Number of books in each category.

SELECT category, COUNT(*) AS books FROM Book GROUP BY category;

SCREENSHOT-



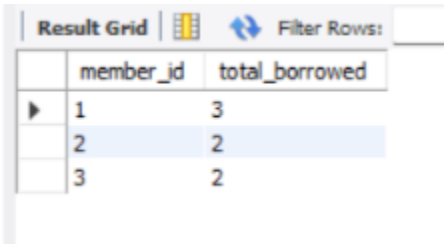
Result Grid | Filter Rows:

category	books
Computer Science	4
Biography	2
Fiction	1
Mathematics	1
Poetry	1
Physics	1

Q9. (HAVING) Members who have borrowed more than one book.

SELECT member_id, COUNT(*) AS total_borrowed FROM Issue
GROUP BY member_id HAVING COUNT(*) > 1;

SCREENSHOT-



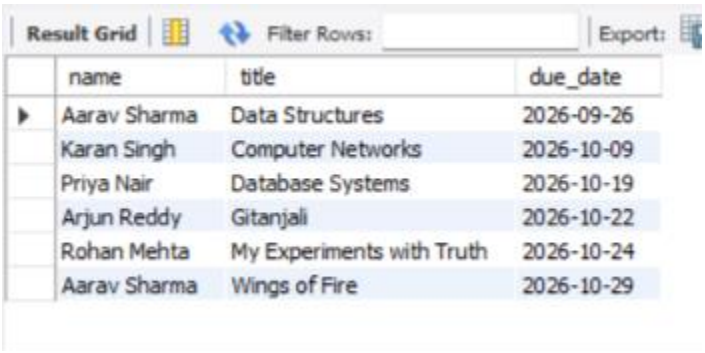
Result Grid | Filter Rows:

member_id	total_borrowed
1	3
2	2
3	2

Q10. (INNER JOIN, 3 tables) Which books are currently not returned, and by whom?

```
SELECT m.name, b.title, i.due_date
FROM Issue i
JOIN Member m ON i.member_id = m.member_id
JOIN Book b ON i.book_id = b.book_id
WHERE i.return_date IS NULL;
```

SCREENSHOT-

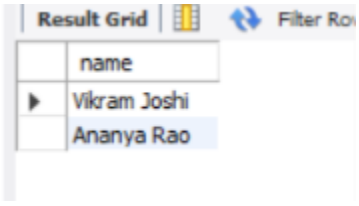

Result Grid | Filter Rows: | Export:

name	title	due_date
Aarav Sharma	Data Structures	2026-09-26
Karan Singh	Computer Networks	2026-10-09
Priya Nair	Database Systems	2026-10-19
Arjun Reddy	Gitanjali	2026-10-22
Rohan Mehta	My Experiments with Truth	2026-10-24
Aarav Sharma	Wings of Fire	2026-10-29

Q11. (LEFT JOIN) Members who have never borrowed a book.

```
SELECT m.name FROM Member m
LEFT JOIN Issue i ON m.member_id = i.member_id
WHERE i.issue_id IS NULL;
```

SCREENSHOT-

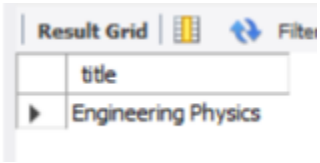

Result Grid | Filter Ro

name
Vikram Joshi
Ananya Rao

Q12. (Subquery with NOT IN) Books that have never been issued.

SELECT title FROM Book
WHERE book_id NOT IN (SELECT book_id FROM Issue);

SCREENSHOT-

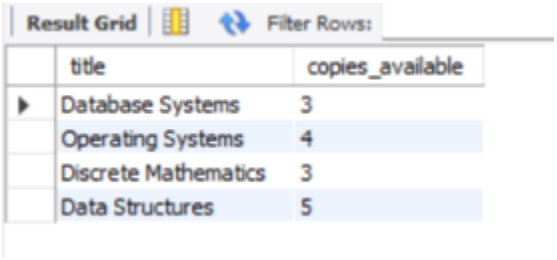

Result Grid | Filte

	title
▶	Engineering Physics

Q13. (Subquery with aggregate) Books with more copies than the average.

SELECT title, copies_available FROM Book
WHERE copies_available > (SELECT AVG(copies_available) FROM Book);

SCREENSHOT-

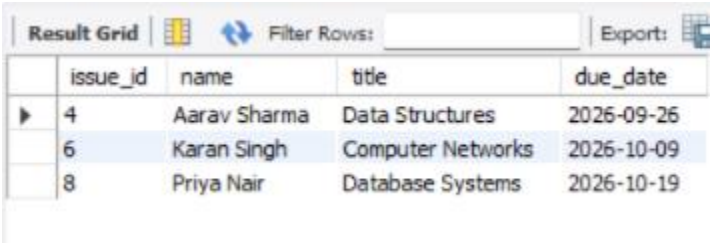

Result Grid | Filter Rows:

	title	copies_available
▶	Database Systems	3
	Operating Systems	4
	Discrete Mathematics	3
	Data Structures	5

Q14. (JOIN + date condition) Overdue books (not returned and due date before 22 Oct 2026).

```
SELECT i.issue_id, m.name, b.title, i.due_date
FROM Issue i
JOIN Member m ON i.member_id = m.member_id
JOIN Book b ON i.book_id = b.book_id
WHERE i.return_date IS NULL AND i.due_date < '2026-10-22';
```

SCREENSHOT-



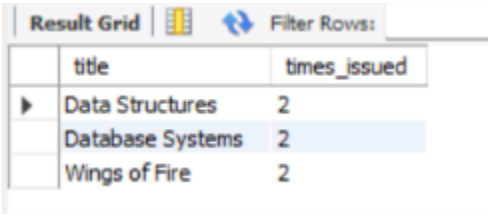
Result Grid | Filter Rows: | Export:

issue_id	name	title	due_date
4	Aarav Sharma	Data Structures	2026-09-26
6	Karan Singh	Computer Networks	2026-10-09
8	Priya Nair	Database Systems	2026-10-19

Q15. (GROUP BY + ORDER BY + LIMIT) Top 3 most issued books.

```
SELECT b.title, COUNT(*) AS times_issued
FROM Issue i JOIN Book b ON i.book_id = b.book_id
GROUP BY b.title
ORDER BY times_issued DESC, b.title
LIMIT 3;
```

SCREENSHOT-



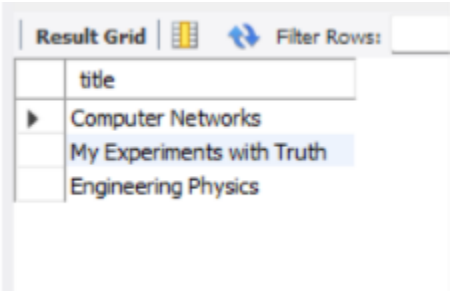
Result Grid | Filter Rows:

title	times_issued
Data Structures	2
Database Systems	2
Wings of Fire	2

Q16. (UNION) Books that need attention: low stock (1 copy or less) or never issued.

SELECT title FROM Book WHERE copies_available <= 1
UNION
SELECT title FROM Book WHERE book_id NOT IN (SELECT book_id FROM Issue);

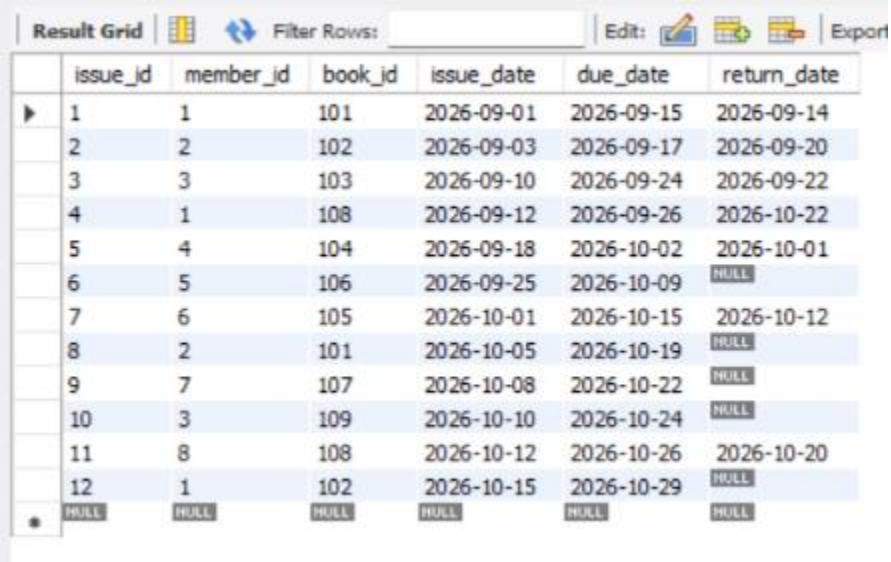
SCREENSHOT-



Q17. (UPDATE) Aarav returns Data Structures today (issue 4).

UPDATE Issue SET return_date = '2026-10-22' WHERE issue_id = 4;
SELECT issue_id, return_date FROM Issue WHERE issue_id = 4;

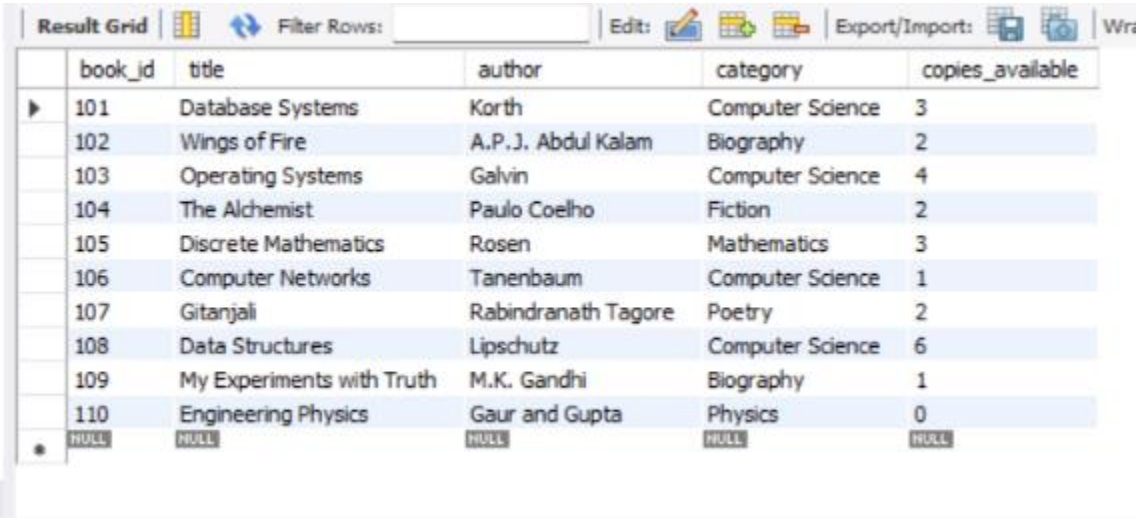
SCREENSHOT-



Q18. (UPDATE with calculation) Add the returned copy back to stock.

UPDATE Book SET copies_available = copies_available + 1 WHERE book_id = 108;
SELECT title, copies_available FROM Book WHERE book_id = 108;

SCREENSHOT-



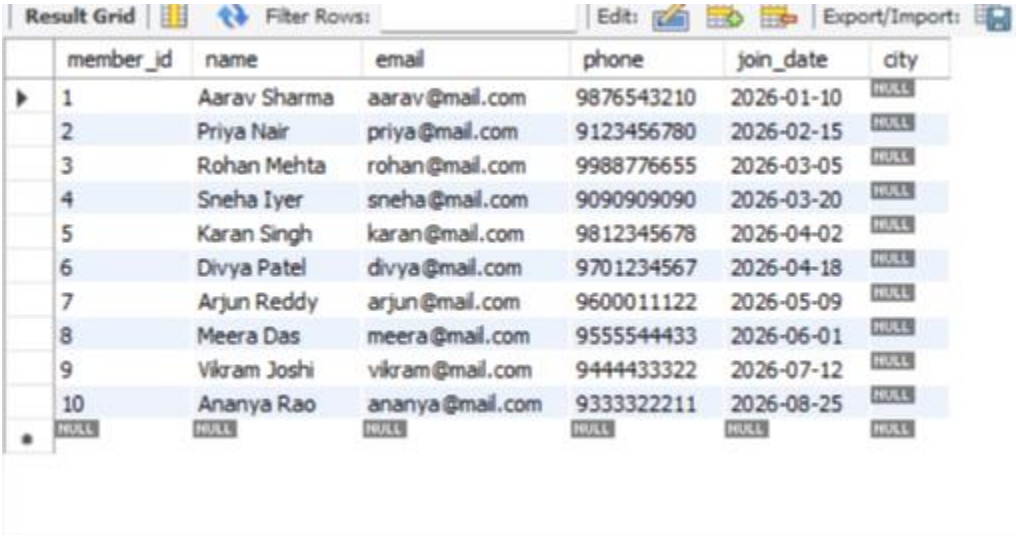
Result Grid | Filter Rows: | Edit: | Export/Import:

book_id	title	author	category	copies_available
101	Database Systems	Korth	Computer Science	3
102	Wings of Fire	A.P.J. Abdul Kalam	Biography	2
103	Operating Systems	Galvin	Computer Science	4
104	The Alchemist	Paulo Coelho	Fiction	2
105	Discrete Mathematics	Rosen	Mathematics	3
106	Computer Networks	Tanenbaum	Computer Science	1
107	Gitanjali	Rabindranath Tagore	Poetry	2
108	Data Structures	Lipschutz	Computer Science	6
109	My Experiments with Truth	M.K. Gandhi	Biography	1
110	Engineering Physics	Gaur and Gupta	Physics	0
NULL	NULL	NULL	NULL	NULL

Q19. (ALTER TABLE) Add a city column to Member.

ALTER TABLE Member ADD COLUMN city VARCHAR(30);

SCREENSHOT-



Result Grid | Filter Rows: | Edit: | Export/Import:

member_id	name	email	phone	join_date	city
1	Aarav Sharma	aarav@mail.com	9876543210	2026-01-10	NULL
2	Priya Nair	priya@mail.com	9123456780	2026-02-15	NULL
3	Rohan Mehta	rohan@mail.com	9988776655	2026-03-05	NULL
4	Sneha Iyer	sneha@mail.com	9090909090	2026-03-20	NULL
5	Karan Singh	karan@mail.com	9812345678	2026-04-02	NULL
6	Divya Patel	divya@mail.com	9701234567	2026-04-18	NULL
7	Arjun Reddy	arjun@mail.com	9600011122	2026-05-09	NULL
8	Meera Das	meera@mail.com	9555544433	2026-06-01	NULL
9	Vikram Joshi	vikram@mail.com	9444433322	2026-07-12	NULL
10	Ananya Rao	ananya@mail.com	9333322211	2026-08-25	NULL
NULL	NULL	NULL	NULL	NULL	NULL

Q20. (VIEW) Create a view of currently borrowed books and display it.

```
CREATE VIEW Currently_Borrowed AS
SELECT m.name, b.title, i.due_date
FROM Issue i
JOIN Member m ON i.member_id = m.member_id
JOIN Book b ON i.book_id = b.book_id
WHERE i.return_date IS NULL;

SELECT * FROM Currently_Borrowed;
```

SCREENSHOT-

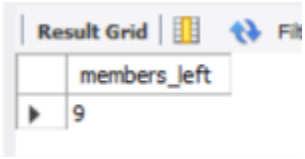


name	title	due_date
Karan Singh	Computer Networks	2026-10-09
Priya Nair	Database Systems	2026-10-19
Arjun Reddy	Gitanjali	2026-10-22
Rohan Mehta	My Experiments with Truth	2026-10-24
Aarav Sharma	Wings of Fire	2026-10-29

Q22. (DELETE) Delete a member who has no issue records (Ananya, id 10).

```
DELETE FROM Member WHERE member_id = 10;
SELECT COUNT(*) AS members_left FROM Member;
```

SCREENSHOT-



members_left
9

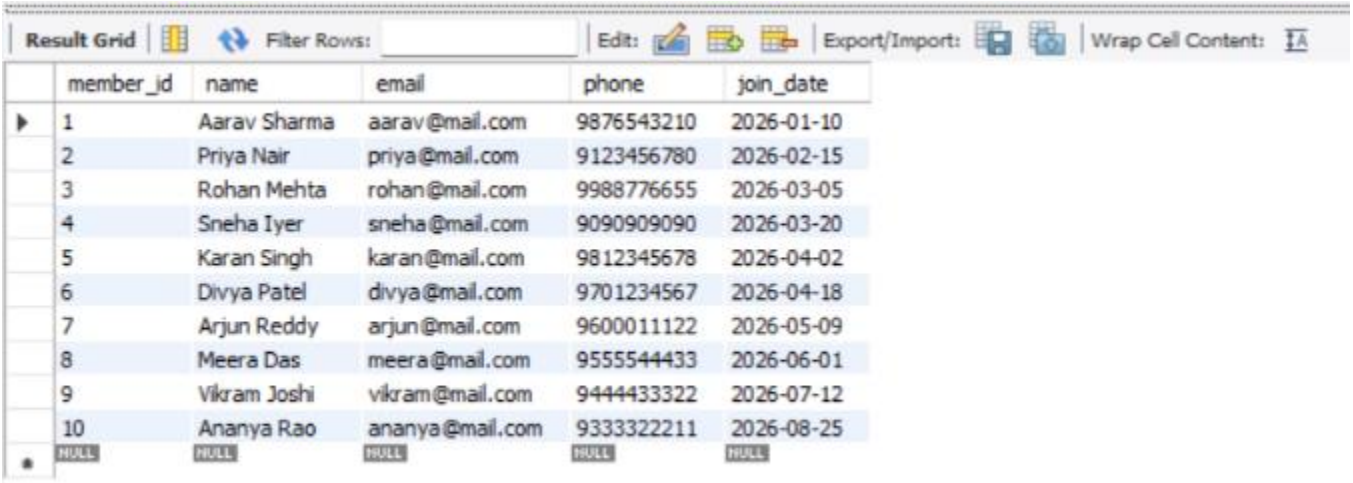
VI. Project demonstration

Tools used:

- SQLite3 as the DBMS (command line or DB Browser for SQLite)
- draw.io for the ER diagram
- Microsoft Word for the report

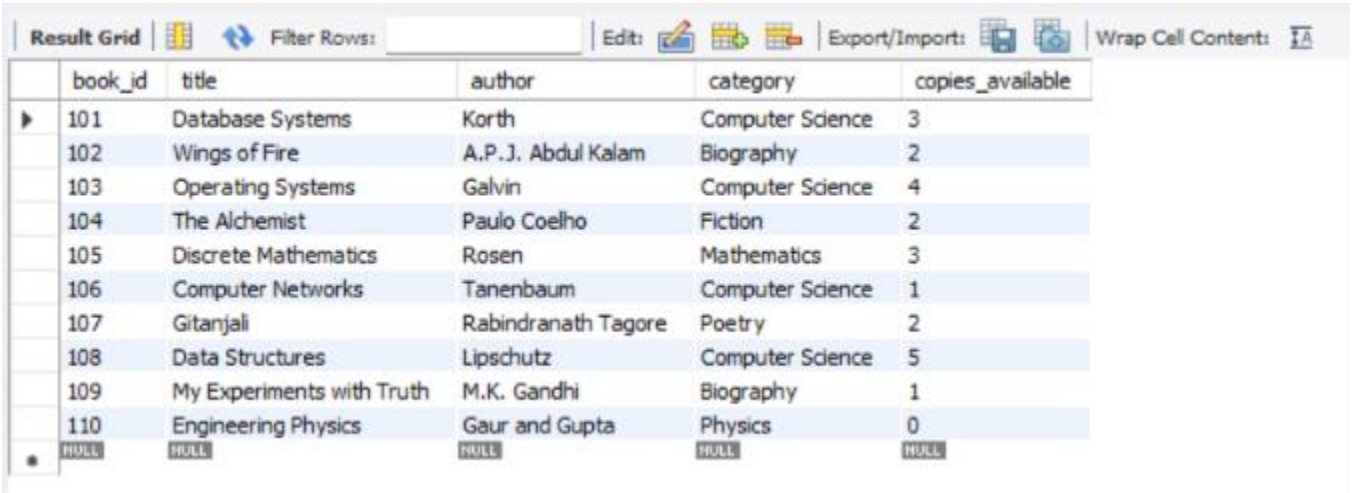
No GUI was built; the project is demonstrated through the SQL command line.

SCREENSHOTS-



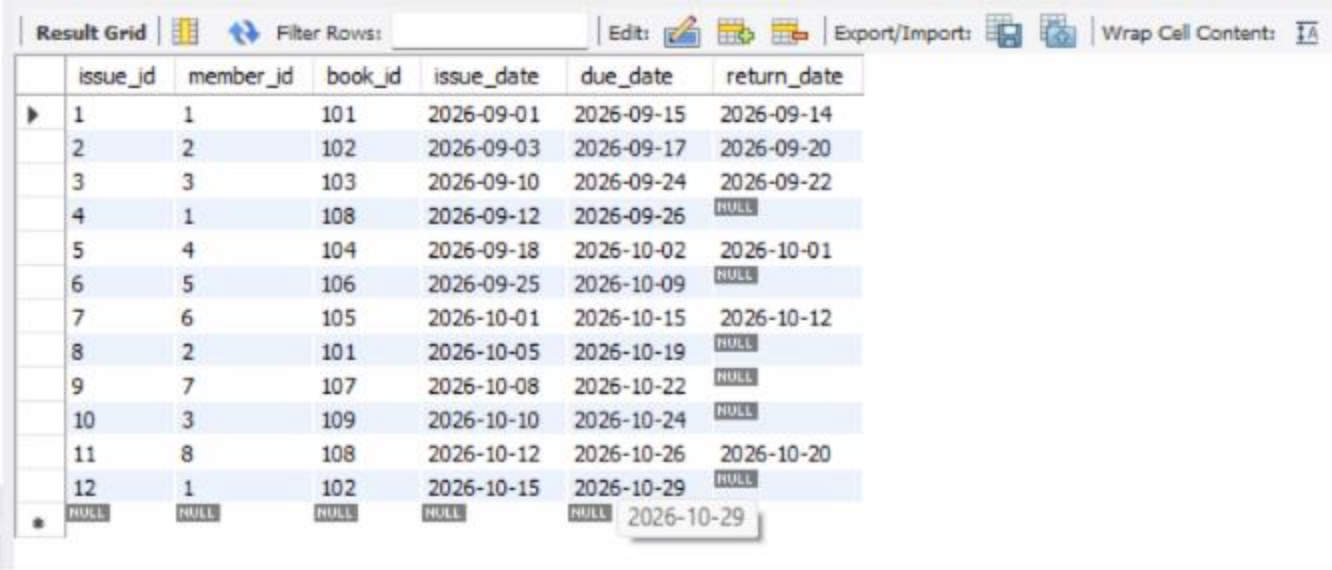
Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

member_id	name	email	phone	join_date
1	Aarav Sharma	aarav@mail.com	9876543210	2026-01-10
2	Priya Nair	priya@mail.com	9123456780	2026-02-15
3	Rohan Mehta	rohan@mail.com	9988776655	2026-03-05
4	Sneha Iyer	sneha@mail.com	9090909090	2026-03-20
5	Karan Singh	karan@mail.com	9812345678	2026-04-02
6	Divya Patel	divya@mail.com	9701234567	2026-04-18
7	Arjun Reddy	arjun@mail.com	9600011122	2026-05-09
8	Meera Das	meera@mail.com	9555544433	2026-06-01
9	Vikram Joshi	vikram@mail.com	9444433322	2026-07-12
10	Ananya Rao	ananya@mail.com	9333322211	2026-08-25
NULL	NULL	NULL	NULL	NULL



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

book_id	title	author	category	copies_available
101	Database Systems	Korth	Computer Science	3
102	Wings of Fire	A.P.J. Abdul Kalam	Biography	2
103	Operating Systems	Galvin	Computer Science	4
104	The Alchemist	Paulo Coelho	Fiction	2
105	Discrete Mathematics	Rosen	Mathematics	3
106	Computer Networks	Tanenbaum	Computer Science	1
107	Gitanjali	Rabindranath Tagore	Poetry	2
108	Data Structures	Lipschutz	Computer Science	5
109	My Experiments with Truth	M.K. Gandhi	Biography	1
110	Engineering Physics	Gaur and Gupta	Physics	0
NULL	NULL	NULL	NULL	NULL



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

issue_id	member_id	book_id	issue_date	due_date	return_date
1	1	101	2026-09-01	2026-09-15	2026-09-14
2	2	102	2026-09-03	2026-09-17	2026-09-20
3	3	103	2026-09-10	2026-09-24	2026-09-22
4	1	108	2026-09-12	2026-09-26	NULL
5	4	104	2026-09-18	2026-10-02	2026-10-01
6	5	106	2026-09-25	2026-10-09	NULL
7	6	105	2026-10-01	2026-10-15	2026-10-12
8	2	101	2026-10-05	2026-10-19	NULL
9	7	107	2026-10-08	2026-10-22	NULL
10	3	109	2026-10-10	2026-10-24	NULL
11	8	108	2026-10-12	2026-10-26	2026-10-20
12	1	102	2026-10-15	2026-10-29	NULL
NULL	NULL	NULL	NULL	NULL	2026-10-29

VII. Self -Learning beyond classroom:

- Practiced extra SQL on free sites such as W3Schools and SQLZoo.
- Learned date functions (julianday) to calculate late days and fines.
- Learned to create **views** and use LEFT JOIN, NOT IN and UNION, which helped answer real library questions.
- Learned to use CHECK, UNIQUE and NOT NULL constraints to keep data correct.
- Learned to draw ER diagrams with draw.io and to use SQLite and DB Browser.

VIII. Learning from the Project

- How to turn a real-life problem into entities, relationships, and tables.
- The difference between primary keys and foreign keys, and how they protect data.
- How a many-to-many relationship is solved with a third table.
- How normalization (1NF to BCNF) removes repeated data.
- How to write 20+ SQL queries from simple SELECT to joins, subqueries and views.
- The project helped me connect classroom theory to a real-world database.

IX. Challenges Faced

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- Deciding which tables were needed and how to connect them.
- Understanding the many-to-many relationship and why Issue is needed.
- Foreign key errors when inserting in the wrong order (Member and Book must be filled before Issue).
- Understanding the difference between 3NF and BCNF.
- Calculating late days, since date functions differ between SQLite and MySQL.

X. Conclusion

This project shows how a database can replace paper records in a library. The three-table design (Member, Book, Issue) stores all the data without repetition and is in BCNF. SQL queries let the librarian quickly see borrowed books, overdue books, popular books, and fines. Constraints such as primary keys, foreign keys, UNIQUE, and CHECK keep the data correct. In future, the project can be extended with fines, tables, librarian login, book reservations, and a simple GUI.